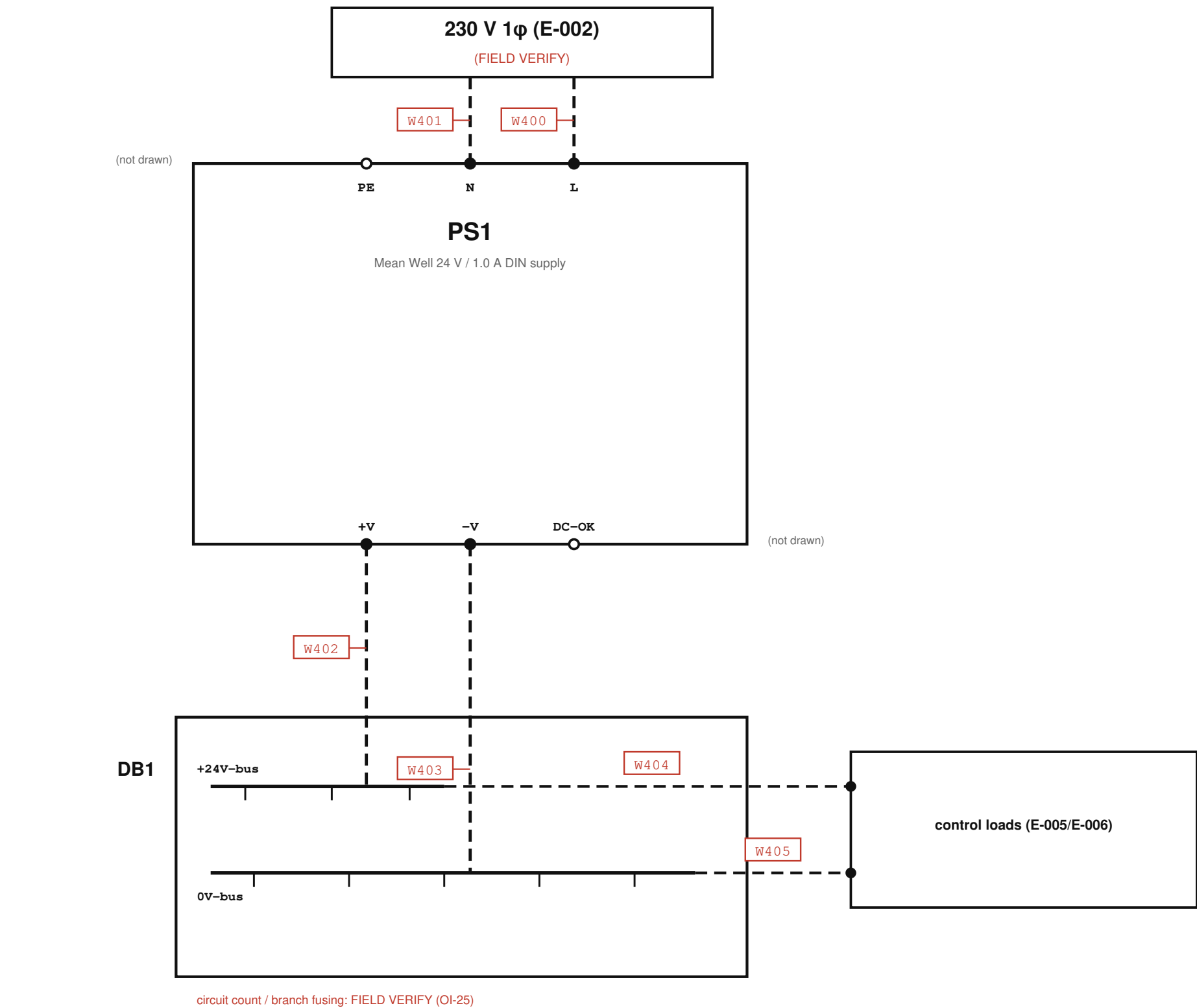


E-004 24 VDC CONTROL POWER

Conv_Simple / CV-101 (garage conveyor) — AC → PS1 (Mean Well 24V/1A) → DB1 +/- distribution block → +24V/0V to E-005, E-006 loads



PS1 is a verified Mean Well DIN-rail supply (24 V / 1.0 A out, 100-240 VAC / 0.55 A in) — nameplate read directly (photo review/photos/wire_2.jpg, 2026-07-11). DB1 polarity (blue vs white = +24V vs 0V), circuit count, PS1 branch fusing, and PS1's AC input source (which leg/circuit feeds L, N) are FIELD VERIFY (OI-25).

SAFETY

- PS1 is fed from AC mains — LOTO + de-energize before opening the panel or metering DB1 (WI p.2 §2).
- Monitored e-stop is NOT a safety stop; de-energizing DB1/PS1 does not remove drive power — see E-003 for the drive supply isolation point.

LEGEND

- VERIFIED (solid)
- FIELD VERIFY (dashed)
- proposed wire number (red = unverified)

CONNECTION TABLE

Wire	From	To	Signal	Type	Status	Notes
W400	230 V 1φ (E-002)	PS1.L	AC line in	power_line	field_verify	
W401	230 V 1φ (E-002)	PS1.N	AC neutral in	power_line	field_verify	
W402	PS1.+V	DB1.+24V-bus	+24 VDC distribution feed	control_24vdc	field_verify	
W403	PS1.-V	DB1.0V-bus	0V distribution return	control_0v	field_verify	
W404	DB1.+24V-bus	control loads (E-005/E-006)	+24V to E-005/E-006 loads	control_24vdc	field_verify	
W405	DB1.0V-bus	control loads (E-005/E-006)	0V to E-005/E-006 loads	control_0v	field_verify	

NOTES

- DB1 is a push-in (WAGO-style), 2-level distribution block fed from PS1's +V/-V output; a bare-copper jumper bridges its top-left terminals in the photo (review/photos/dc_block_full.jpg) — internal jumpering/circuit count not documented.
- PS1's DC-OK terminal (LED/relay status output) and PE input terminal are shown for reference only; neither carries a modeled conductor on this sheet.
- PS1 enclosure/chassis ground and whether DC-0V is bonded to PE anywhere (SELV/PELV scheme vs intentionally floating) is NOT documented — FIELD VERIFY (OI-28).
- +24V / 0V downstream of DB1 feed the E-005 input rail (W24/W0V) and the E-006 output bank 0 feed/return (W600/W609) — same electrical nodes, different sheet.

SOURCES:

photo review/photos/wire_2.jpg (2026-07-11 — Mean Well nameplate read directly)
photo review/photos/dc_block_full.jpg + wire_1.jpg (2026-07-11 — DC distribution block)
review/PHOTO_EVIDENCE_V4.md · review/PHOTO_EVIDENCE_V5.md (OI-25)

24 VDC CONTROL POWER Conv_Simple / CV-101 (garage conveyor)	SHEET E-004
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